first scan the dataset to find the acceptable range for each coordinate.

- Even paying attention to the dataset's boundaries, sampled points could end up in sparse or empty regions, and as such they could later be removed. Instead, by uniformly drawing points from the dataset, the centroids will be close to points in the data and…
- … centroids will be drawn with higher probability in areas with higher density of points.
- Randomly perturbating the points, however, helps avoiding that all the points will be concentrated in the denser areas.

Code in listing 12.2 is also intuitive, though there are a couple of interesting points to raise.

At line #2 we use a generic sampling function that draws n elements from a set without repetition; the details of this method are not important here, and you don't need to worry about it: most programming languages will provide such a function in their core libraries[4], so it's unlikely you will have to implement it[5].

Listing 12.3 classifyPoints

```
function classifyPoints(points, centroids)
                  #1
  clusters ← []
                            #2
  for i in {0, .., |points|-1} do
                   #3
    minDistance ←
inf                                                          #4
```